

Axle Counter Coil Parameter Analysis

This report calculates the primary transmitter coil parameters required to induce a $3\text{ V}_{\text{p-p}}$ voltage on the secondary receiver coil of an axle counter system. Calculations are performed for sinusoidal and square wave excitation in the 10 kHz to 20 kHz range under the assumption of 99.99% magnetic flux loss.

1. Coil Geometry & Assumed Parameters

The secondary receiver coil parameters are defined by the specification. For the primary transmitter coil, we assume a practical air-core geometry of 100 turns with a 100 cm^2 area.

Parameter	Primary Transmitter Coil (\$p\$)	Secondary Receiver Coil (\$s\$)
Number of Turns (\$N\$)	100 (Assumed)	60 (Given)
Coil Area (\$A\$)	$100\text{ cm}^2 = 0.01\text{ m}^2$ (Assumed)	$600\text{ cm}^2 = 0.06\text{ m}^2$ (Given)
Equivalent Radius (\$r\$)	$5.64\text{ cm} = 0.0564\text{ m}$	$13.82\text{ cm} = 0.1382\text{ m}$
Wire Radius (\$a\$)	0.5 mm (Copper, 1 mm diam.)	0.5 mm (Copper, 1 mm diam.)

2. Inductance & Coupling Calculations

Using basic electromagnetic physics, the self-inductance (L) of a multi-turn thin circular loop coil is: $L \approx \mu_0 N^2 r \left(\ln\left(\frac{8r}{a}\right) - 2 \right)$ where $\mu_0 = 4\pi \times 10^{-7}\text{ H/m}$.

For the coupling under 99.99% flux loss ($\text{efficiency } \eta = 10^{-4}$), we analyze **Interpretation A:** The flux linking the secondary is $\Phi_s = 10^{-4} \Phi_p$ (where Φ_p is the flux generated per turn by the primary). This yields a mutual inductance of: $M = 10^{-4} \frac{N_s}{N_p} L_p$

Calculated Values

- Primary Inductance (\$L_p\$):** 3.41 mH
- Secondary Inductance (\$L_s\$):** 3.56 mH
- Mutual Inductance (\$M\$):** $0.2044\text{ }\mu\text{H}$
- Coupling Coefficient (\$k\$):** 5.87×10^{-5}

3. Sine Wave Excitation (10 kHz - 20 kHz)

Under sinusoidal excitation, the target secondary peak-to-peak voltage $V_{s, p-p} = 3 \text{ V}$ ($V_{s, peak} = 1.5 \text{ V}$) requires a primary peak current of: $I_{p, peak} = \frac{V_{s, peak}}{\omega M} = \frac{1.5}{2\pi f M}$

Without matching, driving this current requires massive voltages due to inductive reactance ($X_p = \omega L_p \gg R_p$). By placing a tuning capacitor C_p in series with the primary: $C_p = \frac{1}{\omega^2 L_p}$ the reactive impedance is cancelled, and the required driving voltage drops to $V_p = I_p R_{p, AC}$.

Sinusoidal Parameters (Interpretation A)

Parameter	10 kHz	20 kHz
Skin Depth (δ)	0.652 mm	0.461 mm
AC Resistance ($R_{p, AC}$)	0.764 Ω	1.045 Ω
Required Peak Current ($I_{p, peak}$)	116.79 A	58.39 A
Required RMS Current ($I_{p, rms}$)	82.58 A	41.29 A
Uncompensated Driving Voltage	50.00 kV _{p-p}	50.00 kV _{p-p}
Resonant Series Capacitor (C_p)	74.35 nF	18.59 nF
Resonant Driving Voltage	178.39 V _{p-p}	122.07 V _{p-p}

[!TIP] Resonant tuning reduces the driving voltage by a factor of over **280×** to **400×**, bringing it into the range of standard high-power electronic drivers.

4. Square Wave Excitation (10 kHz - 20 kHz)

Case 1: Square Wave Voltage (Inductance Dominated)

If the primary is driven by a square-wave voltage of amplitude $V_{p, peak}$, the secondary voltage is also a square wave with: $V_{s, p-p} = \frac{M}{L_p} V_{p, p-p} \implies V_{p, p-p} = \frac{L_p}{M} V_{s, p-p} = 50.00 \text{ kV}_{p-p}$ This voltage remains constant over frequency. The primary current is a triangular wave with:

- 10 kHz Peak Current: 183.45 A
- 20 kHz Peak Current: 91.72 A

Case 2: Square Wave Current with $1 \mu\text{s}$ Rise Time (t_r)

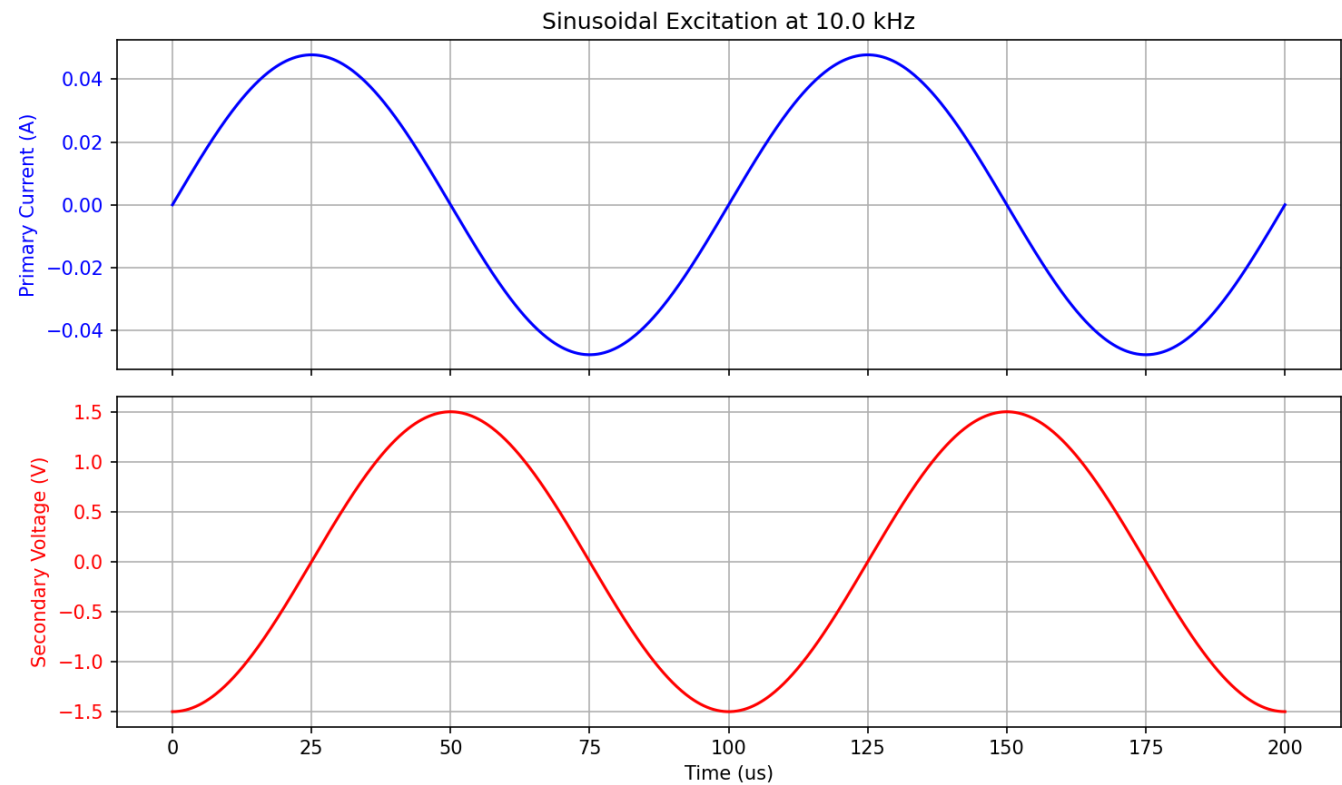
If a square-wave current is driven through the primary, the secondary output consists of short voltage spikes at the transitions. To achieve 3 V_{p-p} spikes:

- Required Peak Current ($I_{p, peak}$): 3.67 A (constant over frequency)
- Required Peak Voltage (during 1 μs transition): $\sim 25,000 \text{ V}$

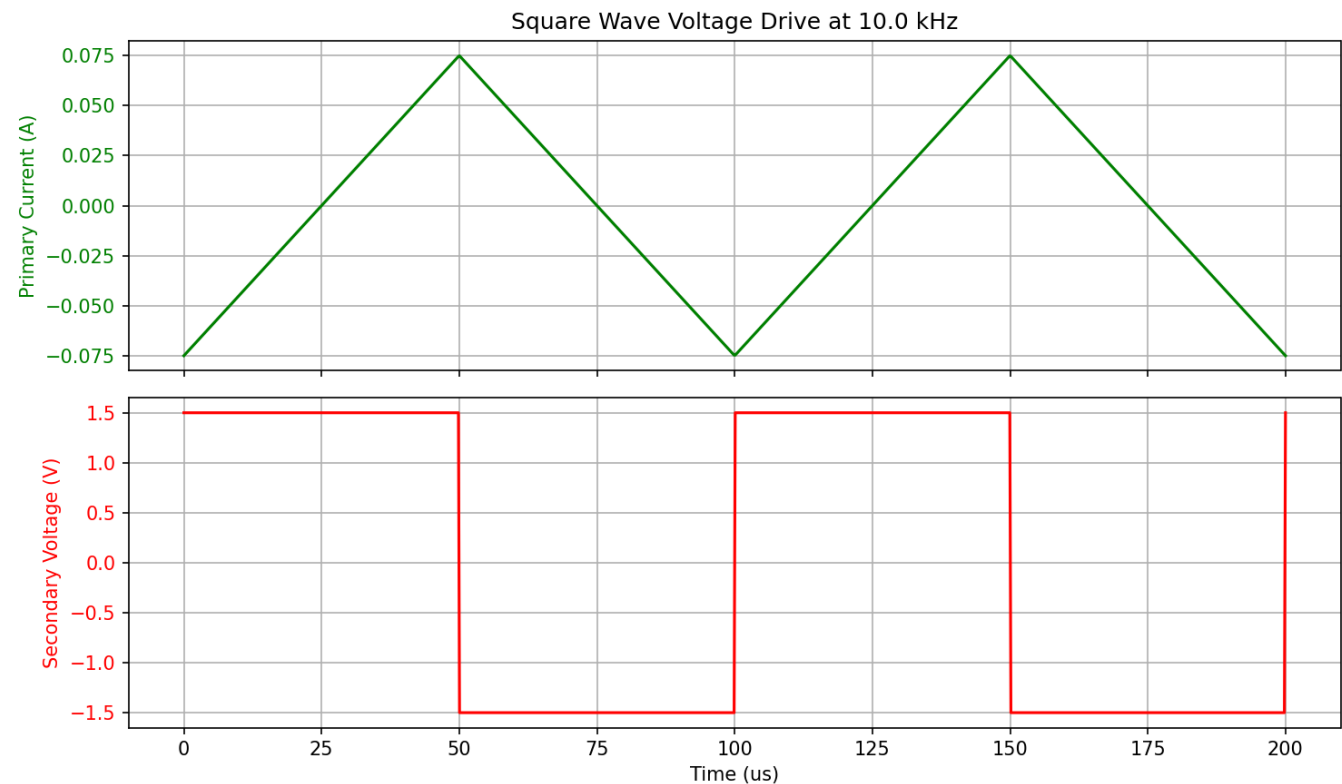
5. Waveform Visualizations

Below are the simulated current and voltage waveforms for 10 kHz excitation, generated in [physics_calculation.ipynb] (file:///c:/Users/rudra/Axle_counter/axlecounter_3dphysics/physics_calculation.ipynb).

Sinusoidal Drive (10 kHz)



Square Wave Voltage Drive (10 kHz)



6. What's Missing? Key Engineering Questions

Basic air-core electromagnetic equations provide a theoretical baseline, but real-world axle counter sensors (e.g., Frauscher, CEL) differ in several critical ways. Please consider the following questions:

1. Ferrite Cores:

- *Are you using air-core coils or wound ferrite cores (e.g., pot or U-cores)?*
- *Why it matters:* High-permeability ferrite cores focus the magnetic flux, reducing flux loss from 99.99% to a much lower value. This reduces the required primary current and voltage by orders of magnitude.

2. The Influence of the Steel Rail:

- *Is the sensor mounted directly onto a steel rail web/head?*
- *Why it matters:* The ferromagnetic steel rail acts as a massive magnetic shunt and introduces significant eddy-current losses. This drastically alters the baseline coupling and creates phase shifts.

3. Resonant Secondary (Rx Tuning):

- *Are you planning to tune the secondary coil with a parallel or series capacitor?*
- *Why it matters:* Resonance on the receiver side boosts the induced voltage by the quality factor ($Q_s \approx 30 - 80$). This means you would only need $\sim 0.05 \text{ V}$ open-circuit voltage from the coupling itself to get a 3 V resonant output, reducing transmitter power by $50\times$.

4. Preamplifier Load Impedance:

- *What is the input impedance of the receiver's preamplifier?*
- *Why it matters:* A low input impedance loads the secondary coil, reducing its voltage and quality factor (Q), which must be factored into the electrical model.